# 📘 Azure + Azure DevOps (ADO)

**(AWS Correlated, Practical, Security + Networking Focus)**

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## 🧠 1. Azure Structure (AWS Mapping)

### Tenant → AWS Organization

- Represents your company's identity in Azure.
- Manages users, groups, authentication, and directory-wide policies.
- **Use case:** Single tenant per company; multiple subscriptions live under it.

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### Subscription → AWS Account

- Billing and access boundary, like an AWS account.
- Used to separate environments: Dev, Test, Prod.
- **Example:** Dev team uses one subscription, Prod team another to prevent accidental downtime.

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### Resource Group → AWS CloudFormation Stack / Tag Grouping

- Logical container for related resources (VMs, Storage, Databases).
- Lifecycle management: deleting the RG deletes all resources inside.
- **Best practice:** Create separate RGs for each application or environment.

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## ⚙️ 2. Compute Services (AWS Mapping)

### Virtual Machine (VM) → AWS EC2

- Full OS control, custom setups, or legacy apps.
- Ideal when server management is required.
- Daily DevOps: monitor health, patch OS, scale resources.

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## App Service → AWS Elastic Beanstalk / Lambda (PaaS)

- Managed platform for web apps, APIs, and microservices.
- Auto scaling, fast deployment, no server management.
- **Example:** Deploy Node.js API quickly without managing OS.

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## AKS → AWS EKS

- Managed Kubernetes service for containerized apps.
- Auto-scaling, monitoring, and integrated with DevOps pipelines.
- **Daily task:** Monitor pods, scale nodes, troubleshoot container crashes.

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# 💾 3. Storage Services (AWS Mapping)

## Blob Storage → AWS S3

- Object storage for logs, images, videos, backups.
- Practical: store app logs, archive old data.

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## File Storage → AWS EFS

- Shared network-attached file system.
- Use case: multiple servers sharing configurations or reports.

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## Managed Disks → AWS EBS

- Persistent storage for VMs.
- Daily task: resize, snapshot, attach/detach disks.

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# 🌐 4. Networking (AWS Mapping)

## Virtual Network (VNet) → AWS VPC

- Private network for Azure resources, isolates workloads.

- Can span regions with peering, supports hybrid connections.
- **Practical:** Separate prod and dev workloads within same subscription.

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## Subnet → AWS Subnet

- Segments within VNet for traffic control and isolation.
- **Example:** Separate frontend, backend, database tiers.

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## Network Security Group (NSG) → AWS Security Group

- Firewall for inbound/outbound traffic.
- Daily task: restrict ports, allow specific IPs, secure access.

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## Load Balancer → AWS ELB

- Distributes incoming traffic across multiple resources.
- Ensures high availability and redundancy.
- **Use case:** Scale web app to handle high traffic.

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## VPN Gateway → AWS VPN / Direct Connect

- Connects on-premise network to Azure securely.
- **Use case:** Hybrid infrastructure, secure connections for enterprise apps.

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# 🔐 5. Security Services (AWS Correlation)

## Azure AD → AWS IAM

- Identity and access management for users, groups, and applications.
- **Daily task:** Manage users, enforce multi-factor authentication, SSO.

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## RBAC → AWS IAM Roles

- Assign minimum necessary permissions per resource (Owner, Contributor, Reader).

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- **Use case:** Developers get Contributor on Dev RG, read-only on Prod RG.

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## Key Vault → AWS Secrets Manager / KMS

- Secure storage for passwords, certificates, API keys.
- Avoid storing sensitive information in code or configs.

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## Defender for Cloud → AWS GuardDuty / Security Hub

- Security monitoring, threat detection, vulnerability assessment.
- **Daily task:** Monitor alerts, remediate misconfigurations, track security compliance.

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## Azure Firewall → AWS Network Firewall

- Centralized security for VNets.
- Controls traffic across multiple subnets and connections.

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## DDoS Protection → AWS Shield

- Protects apps from denial-of-service attacks.
- Ensures high availability during attacks.

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# 🧰 6. Azure DevOps (ADO) – AWS Correlation

## What is ADO?

- End-to-end DevOps platform integrating code, work tracking, build, and deployment.
- Similar to AWS CodeCommit, CodePipeline, CodeBuild, and Jira.

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## Boards → Jira / AWS CodeStar Issues

- Track tasks, bugs, features.

- Supports Scrum/Kanban; manage sprints and releases.
- **Practical:** Assign tasks, track progress, daily updates.

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## Repos → AWS CodeCommit / GitHub

- Git-based code repositories for version control.
- Supports branching, pull requests, and collaboration.

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## Pipelines → AWS CodePipeline

- Automates building, testing, deploying applications.
- **Daily DevOps:** Monitor builds, fix failed jobs, trigger deployments.

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## Artifacts → AWS CodeArtifact

- Stores packages and dependencies for reuse across apps and teams.
- Example: Share NPM, NuGet, Maven packages in enterprise apps.

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## Test Plans → AWS CodeBuild / Manual Testing

- Track test cases, automated or manual tests.
- Ensures quality before deployment.

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# 📊 7. Monitoring (AWS CloudWatch Correlation)

## Azure Monitor → AWS CloudWatch

- Collect metrics and logs from apps and infrastructure.
- Create dashboards and alerts for thresholds like CPU, memory, and downtime.

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## Log Analytics → CloudWatch Logs Insights

- Query and analyze log data using KQL.
- Daily task: debug errors, analyze failed jobs, troubleshoot incidents.

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## Application Insights → AWS X-Ray

- Monitors application performance and usage patterns.
- **Practical:** Detect slow APIs, exceptions, and request failures.

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# 💰 8. Cost Management (AWS Cost Explorer)

- Track and optimize cloud spending per subscription/resource group.
- Daily DevOps: identify idle or over-provisioned resources, resize, or delete.
- **Example:** Auto-stop Dev VMs outside working hours to save cost.

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# 🧪 9. Daily DevOps Tasks

- **Monitoring:** Check dashboards, alerts, performance metrics.
- **Debugging:** Troubleshoot application or infrastructure issues.
- **Deployment:** Release updates to App Service or AKS, rollback if needed.
- **Security:** Manage RBAC, NSGs, Key Vault, Defender alerts.
- **Collaboration:** Use Boards, Repos, Artifacts to track work.

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# 🎯 10. Interview Preparation (AWS → Azure Mapping)

- Subscription vs Resource Group → AWS account vs CloudFormation stack.
- VM vs AKS → EC2 vs EKS.
- Monitoring → Azure Monitor / Log Analytics vs CloudWatch.
- Security → Azure AD/RBAC/Key Vault vs IAM/Secrets Manager.

**Scenario Questions:**

- App down → check monitoring, logs, infra, deploy fix.
- High cloud bill → analyze resources, optimize, remove idle services.
- Security incident → restrict access, rotate secrets, fix misconfigurations.

# 🌐 AWS → Azure → Azure DevOps (ADO) Cheat Sheet

| AWS Service / Concept | Azure Equivalent | Daily DevOps / Practical Use |
|---|---|---|
| AWS Organization | Tenant | Manage company-wide users, groups, authentication, and directory-wide policies |
| AWS Account | Subscription | Separate environments (Dev/Test/Prod), billing, access control |
| CloudFormation Stack / Tags | Resource Group | Deploy and manage related resources as a unit; lifecycle management |
| EC2 | Virtual Machine (VM) | Monitor VM health, patch OS, scale resources, handle legacy apps |
| Elastic Beanstalk / Lambda | App Service | Deploy web apps/APIs, auto scaling, no server management |
| EKS | AKS | Container orchestration, monitor pods, scale nodes, troubleshoot containers |
| S3 | Blob Storage | Store logs, backups, images, videos |
| EFS | File Storage | Shared file access across servers or apps |
| EBS | Managed Disks | Persistent VM storage, snapshots, resize disks |
| VPC | Virtual Network (VNet) | Isolate workloads, control routing, enable hybrid connections |
| Subnet | Subnet | Organize resources by tier (frontend/backend/db), traffic control |
| Security Group | NSG | Control inbound/outbound traffic, secure resources |
| ELB | Load Balancer | Distribute traffic, ensure high availability |
| VPN / Direct Connect | VPN Gateway | Secure hybrid network connection |
| IAM | Azure AD | Identity & access management, SSO, multi-factor authentication |
| IAM Roles | RBAC | Assign minimum privilege access (Owner, Contributor, Reader) |
| Secrets Manager / KMS | Key Vault | Securely store secrets, API keys, certificates |
| GuardDuty / Security Hub | Defender for Cloud | Security monitoring, threat detection, vulnerability assessment |
| CloudWatch | Azure Monitor | Collect metrics and logs, create dashboards and alerts |
| CloudWatch Logs Insights | Log Analytics | Query logs using KQL, debug issues, troubleshoot failures |

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| AWS Service / Concept | Azure Equivalent | Daily DevOps / Practical Use |
|---|---|---|
| X-Ray | Application Insights | Monitor application performance, response times, failures |
| CodeCommit / GitHub | Repos | Version control, branching, pull requests, team collaboration |
| CodePipeline | Pipelines | Automate build, test, deployment workflows |
| CodeArtifact | Artifacts | Store and share packages/libraries across teams |
| CodeBuild / Test | Test Plans | Track automated or manual tests, ensure quality before deployment |

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